

Agent

RL Agent (PyTorch)

Wrapper stack

actions
observation + mask, rewards, done, info

StatsRecorder → FrameStack →
ReservedPositionObs → SymmetryAugmentation

Python layer

actions
observation + mask, rewards, done, info

PaddedMicroRTSGridModeVecEnv
multi-map, switch_map()

MicroRTSGridModeVecEnv
observation encoding, action dispatch

MicroRTSBotVecEnv
bot vs bot (tournaments)

BaseMicroRTSVecEnv
JVM bootstrap, shared interface

actions
raw observation + raw mask, raw rewards, done, info

JPytype / JNI bridge

bridge.jar → microrts.jar → Bot JARs

Java layer

JNIGridnetVecClient — N parallel games

JNIGridnetClient-SelfPlay
agent vs agent

JNIGridnetClient
agent vs bot

JNIBotClient
bot vs bot

- GameStateWrapper
- FilteredMaskClient
- RewardFunctions

MicroRTS Engine